Market Regime Detection: A Hidden Markov Model Approach

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Abstract

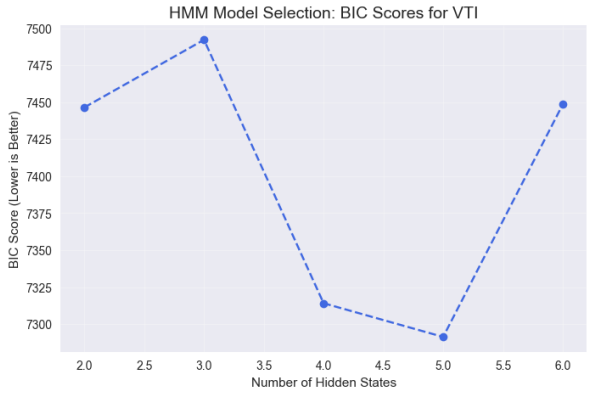
The goal of this project was to use a Gaussian Hidden Markov Model (HMM) to identify market regimes and investigate how different sectors (Energy and Technology) perform under each total market regime. Overall, the project found that sector diversification (specifically between Technology and Energy) is a function of the market regime, effectively "breaking" during crisis states and moving together in euphoric states (a phenomenon known as correlational convergence).

Methodology & Model Selection

Like most projects working with data, the first portion of the project was allocated to ingesting and preprocessing the data. Using yfinance, ticker data is extracted into pandas DataFrames and filtered into 2 columns: date and close price. VTI was used as the global state (or total market data) for this project while XLK and XLE were used to represent the Technology and Energy sector respectively. After retrieving the dataset, the final part was combining the data for each ticker into a singular DataFrame, which was then used to return a separate DataFrame with each ticker and its daily log returns.

In the exploratory data analysis (EDA) portion, VTI's daily log returns and distribution were plotted to observe volatility clustering and excess kurtosis, justifying the use of a Markovian approach due to its memoryless nature. At this point, a question remained: how many states should be classified and observed? This is where the HMM steps in and, under the Bayesian Information Criterion (BIC), the optimal number of states was found: 5. In short, the BIC passes the log-likelihood (or explainability) of the model and adds a penalty as more parameters are added, optimizing the model to not underfit or overfit.

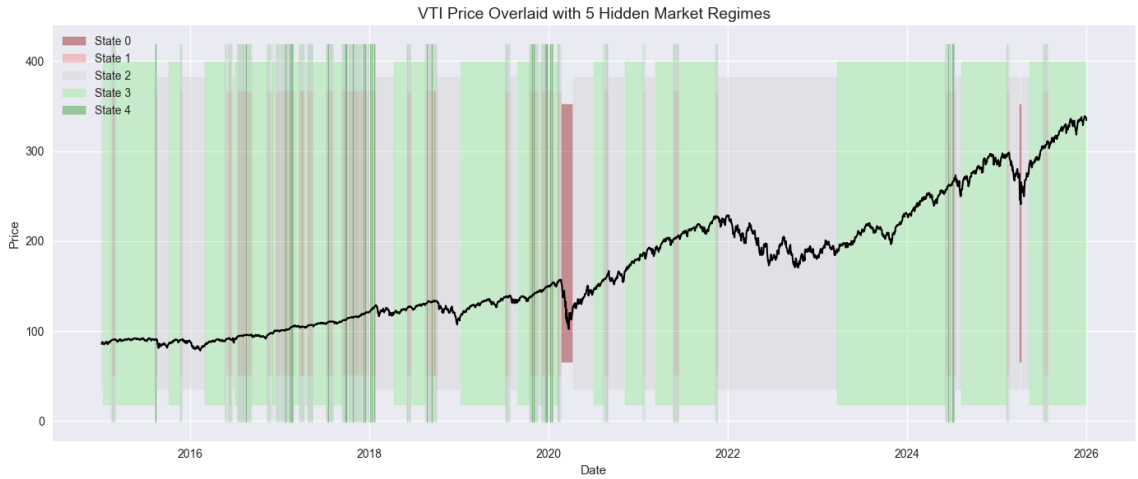
The 5 Phases of the Market



--- VTI 5-State Regime Signatures ---

State	Mean Return (%)	Volatility (%)
0	-0.623265	4.895283
1	-0.110545	0.282572
2	-0.059434	1.464695
3	0.081926	0.809463
4	0.328202	0.375687

The images above show how the BIC identified 5 states as the optimal balance between accuracy and complexity, largely preventing underfitting or overfitting. Undergoing this process was important as many would have expected the market to perhaps only have 3 states (Bull, bear, and neutral). However, as shown in the image on the right, the HMM was able to identify 5 regimes where the middle 3 states serve as transitional micro-regimes. Sorted and labeled based on mean returns, State 0 represents the severe bear state with massively negative returns and high volatility while State 4 represents the high-conviction bull state with great returns and very low volatility.



Once the states were characterized and uncovered, it was time to plot and overlay the identified market regimes with the price of VTI over the entire period. The main takeaway from the plot above is

the clustering of regimes over long periods, confirming the presence of regime inertia. Regime inertia refers to the clustering of the states and shows that the HMM is detecting significant structural shifts in market dynamics rather than random noise. Additionally, this part acted as a sanity check to see if the model captured known downturns or upswings like the March 2020 COVID-19 crash. Ultimately, the clustering validated that market states have a kind of memory, highlighting how once the market enters a state of panic of growth, it tends to stay there.

The Diversification Paradox

The real takeaways from this study come from the above comparison between the inter-sector and total market correlation values within each identified regime. The image on the left identifies XLK and VTI with a correlation factor of 0.92 and XLE and VTI with a correlation factor of 0.64. This serves as a baseline/average correlation over the entirety of the 11-year period. However, these values serve as a mask, where most of the trading and insight comes from the microstructure and transactions that occur daily. As shown above, in State 0 (Severe Bear), XLK and VTI have a very strong correlation factor of 0.98 and XLE and VTI have a strong correlation factor of 0.89. These correlations contrast greatly with the 0.67 between XLK and VTI and the 0.41 between XLE and VTI in State 4. As observed, in these different states, the correlation/movement of these assets with the total market varies greatly. In the severe bearish regime (State 0), energy and technology seemingly move in the same direction, stripping away any hopes of portfolio diversification. In the highly bullish regime (State 4), energy and technology have a negligible correlation, allowing traders to utilize these two sectors as a form of diversification.

However, there are some drawbacks to analyzing and strategizing using this HMM. For one, the model is fundamentally lagging. Due to its Markovian properties, the model requires a spike in volatility to occur before it can reclassify the state, being a few steps behind regime shifts. With that being said, the HMM identified relationships between sectors under different market environments and gave a deeper insight into how traders could navigate portfolio diversification. These takeaways indicate that diversification with energy and technology acts as a "fair-weather friend" that exists when you don't need it and disappears exactly when the market crashes. Ultimately, the data supports the idea of correlational convergence: a phenomenon where two or more assets tend to increase in correlation during period of market stress or volatility (Sumnicht).

References

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